Status	Finished
Started	Thursday, 1 October 2026, 4:07 PM
Completed	Thursday, 1 October 2026, 4:10 PM
Duration	3 mins 10 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 94

b. 90 and 99 ✓

c. 89 and 94

d. 89 and 99

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

Number values should only be present

b. There must be a mechanism to delete and/or insert elements in the list ✓

c. There should be direct access to the middle element in any sublist

d. The list must be sorted

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓

b. Linear search divides the list into halves.

c. Linear search requires the list to be sorted.

d. Linear search is more efficient than binary search on large lists.

Question **4**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search stops
- b. The search continues from the beginning
- c. The list is sorted
- d. The search continues in the left or right sublist ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Linear search ✓
- b. Jump search
- c. Hashing
- d. Binary search

Question **6**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- a. The search continues in the left sublist
- b. The search ends successfully ✓
- c. The list is sorted
- d. The search continues in the right sublist

Question **7**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 89 and 94
- b. 90 and 100
- c. 90 and 99 ✓
- d. 94 and 99

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly

b. Binary search algorithm is not efficient when the data elements more than 1500 ✓

c. Must use a sorted array

d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Checking each element sequentially ✓

b. Skipping every second element

c. Sorting the list before searching

d. Dividing the list in half repeatedly

Question **10**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1

b. log n

c. n/2

d. n ✓

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists

b. It is slower than linear search for small lists

c. It requires the list to be sorted ✓

d. It does not work with negative numbers

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Hashing
- b. Linear search ✓
- c. Dynamic search
- d. Binary search

Question **13**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 500
- b. 10 ✓
- c. 100
- d. 1000

Question **14**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Sequential search
- b. Linear search
- c. Binary search ✓
- d. Hash search

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Merge search
- b. Bubble search
- c. Quick search
- d. Linear search ✓